EXASOL OPEN BUILD CHALLENGE | TRACK 01: MONEY, RISK & REGULATION

CloseLoop: In-Database Reconciliation Engine

Compressing 7-Day Month-End Close into 4 Hours on Exasol Personal MPP

Team CloseLoop: Sneha A. & Sanjeev V.M. | MIT License | Built Exclusively for Exasol Personal

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The Problem: Why Month-End Close Consumes a Full Week

01. The 'Excel VLOOKUP' Bottleneck

Banks process millions of daily transactions across Core GL, Payment Gateways, and Bank feeds.

Finance ops teams dump CSVs into Excel or legacy row-based databases that choke on volume, memory, and multi-field non-exact joins.

02. The Residual Break Spiral

Exact matching fails on ~40% of records due to real-world friction:

- Settlement calendar lag (T+1..T+3)
- Gateway MDR fee cuts (1.5-2.5%)
- Reference typos & OCR mutations
- Many-to-one batch settlements

03. Compounding Regulatory Risk

Unresolved breaks sit in Suspense Accounts for months.

Under statutory audit & RBI guidelines, aged suspense balances trigger mandatory capital provisioning, audit qualifications, and massive analyst overtime.

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Who is Affected: The Human Cost in Finance Operations

Financial Controller / CFO

Pain Point: Stuck with 7-day delayed MIS reporting, Board pressure, and compliance risk of unresolved clearing balances exceeding regulatory audit thresholds.

CloseLoop Impact: Compress close to < 4 hours with certified zero-suspense sign-off.

Reconciliation Operations Lead

Pain Point: Manages 20+ analysts working frantic 14-hour weekend shifts copying and pasting spreadsheets to clear millions of broken transaction lines.

CloseLoop Impact: 91%+ first-pass auto-clear rate; eliminate manual spreadsheet joins.

Statutory / RBI Internal Auditor

Pain Point: Faced with fragmented audit trails, untracked spreadsheet adjustments, and unexplainable journal vouchers dumped into suspense accounts.

CloseLoop Impact: 100% immutable append-only SQL audit ledger with rule & actor provenance.

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CloseLoop Architecture: 100% In-Database Set-Based Execution

1. Ingestion

pyexasol bulk load streams Core Ledger, Payment Gateway JSON/CSV, and Bank feeds directly into Exasol staging tables.

2. Normalisation

Set-based SQL cleans noisy tokens ('UTR-', 'REF#', spaces), aligns ISO dates, and standardizes signed cash flows.

3. 4-Tier Funnel

T1: Exact Match (100%)
T2: Lag & Fees (96%)
T3: Levenshtein Fuzzy (88%)
T4: Batch Settlements (94%)

4. Classification

Classifies residuals into 6 categories: Timing, Fee, Duplicate, Transposition, Missing GL, Structuring.

5. Human Review

Ranked review queue with explainable JSON evidence cards and immutable append-only audit trail.

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Why Exasol: Columnar In-Memory MPP vs Python/Pandas & Legacy DBs

Legacy Python / Pandas

- Single-threaded GIL bottleneck
- Copies entire dataset into RAM (10x bloat)
- Cartesian joins choke at 100k rows
- Crashes with OOM at 10M rows
- Zero ACID transaction or audit guarantees

Legacy Row DBs (Postgres/Oracle)

- Row-oriented disk I/O thrashes on billion-cell joins
- Table lock contention slows bulk inserts
- Complex fuzzy indexing requires third-party plugins
- Scaling requires heavy sharding & manual DBA tuning

Exasol Personal (CloseLoop Engine)

- In-Memory Columnar MPP: 10M rows joined in seconds
- Hardware-accelerated vectorization & JIT compilation
- Built-in EDIT_DISTANCE & Python Scalar UDFs
- Lock-free ACID transactions & bulk HTTP streaming
- Free personal edition scales seamlessly to cluster

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The CloseLoop Analyst UX: High-Velocity Review & Triage

Executive KPI Waterfall

Real-time metrics tracking First-Pass Auto-Clear Rate (90.92%), Baseline Lift (+29.43%), Suspense Float, and Close Cycle compression from 7 days to 4 hours.

Tier-by-Tier SQL Execution

Visual telemetry showing set-based SQL queries running directly inside Exasol: T1 Exact -> T2 Rules -> T3 Fuzzy -> T4 Batch. Complete execution under 2 seconds.

Ranked Human Review Queue

Residual breaks ranked by confidence score. Analysts inspect candidate diffs, fee deductions, and approve/reject with 1-click maker-checker sign-off.

Append-Only Audit Trail

Complete event provenance tracking every automated tier match, break categorization, and manual analyst decision with immutable regulatory timestamps.

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Empirical Results: Scale Benchmark & Ground Truth Evaluation

100.0%
Match Precision
Zero false-positives

99.67%
Ground Truth Recall
Covers true pairs

90.92%
Auto-Clear Rate
+29.4% baseline lift

19.2x
Speedup vs Pandas
At 1 Million rows

DATASET VOLUME	EXASOL PERSONAL	PYTHON / PANDAS	SPEEDUP	AUTO-CLEAR
10,000 Rows	**1.69 seconds**	0.16 seconds	**Baseline (overhead bound)**	90.92%
100,000 Rows	**4.21 seconds**	18.74 seconds	**4.4x faster**	91.15%
1,000,000 Rows	**14.82 seconds**	284.10s (4.7 min)	**19.2x faster**	91.40%
10,000,000 Rows	**78.40s (1.3 min)**	Process Killed (OOM)	**Infinity (MPP)**	91.50%

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Business & Financial Impact: Compressing Close and Slashing Risk

95% Close Cycle Compression

5-7 business days reduced to under 4 hours. Accelerates management reporting, eliminates analyst weekend shifts, and allows timely Board disclosures.

₹12-25 Cr Capital Unlocked

Auto-clearing 91%+ of breaks prevents lingering float in suspense accounts, freeing up liquidity for overnight reverse repo / yield generation.

Zero Audit Qualifications

100% immutable append-only event trail ensures total compliance with statutory audit standards and RBI internal control recordkeeping directives.

Fraud & AML Structuring Defense

In-database heuristics instantly flag structuring patterns (e.g., ₹49,999 cash deposits) and duplicate postings that hide inside manual spreadsheets.

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Production Roadmap: Moving from Challenge Build to Bank Deployment

Phase 1: Hackathon Core (Delivered)

- End-to-end T1-T4 matching inside Exasol Personal
- Synthetic data generator with ground truth
- Review queue with confidence & explainable JSON
- Append-only audit trail & Streamlit console

Phase 2: Bank Integration (Q1 2027)

- Direct SFTP / Kafka ingestion connectors for Finacle, Flexcube, and Razorpay/PineLabs
- Granular Role-Based Access Control (RBAC) & SAML SSO
- Maker-Checker 4-eye workflow for manual overrides

Phase 3: Autonomous Intelligence (Q2 2027)

- LLM-assisted residual break summarizer (explanatory only)
- Adaptive fee envelope estimation per gateway MID
- Cross-entity multi-currency FX revaluation matching

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Summary & Open-Source Artifacts: Complete & Reproducible

Team CloseLoop (Track 01)

Sneha A. & Sanjeev V.M. | Full-stack financial reconciliation on Exasol Personal.

Public GitHub Repository

github.com/SnehaAnand21/exasol-closeloop | MIT License | Clean git history created during challenge window.

Pure In-Database Core

100% of matching, joining, windowing, and scoring runs inside Exasol Personal set-based SQL.

1-Command Reproduction

'make demo' or 'python app/standalone_server.py' delivers a working dashboard from clone in under 60s.